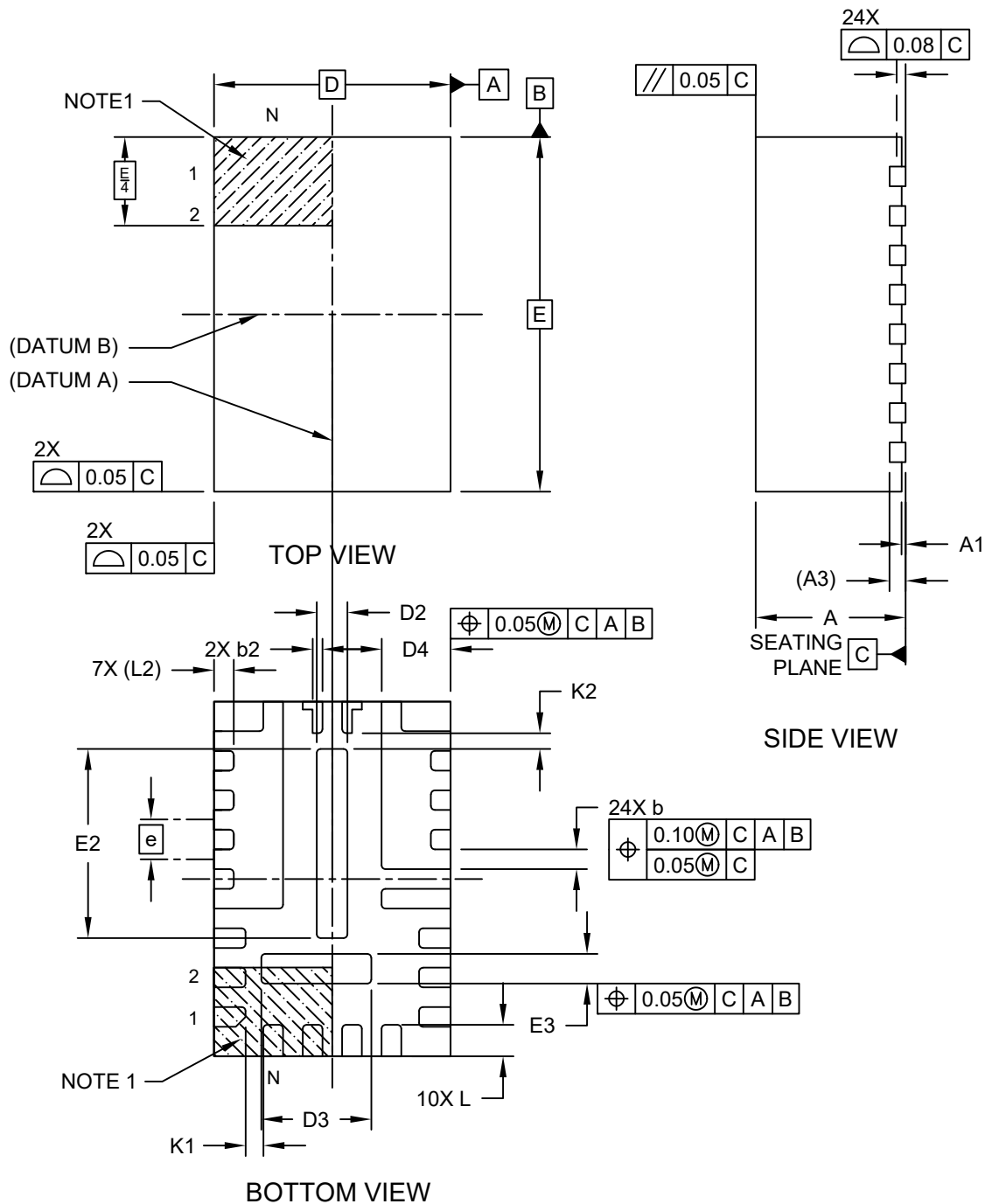


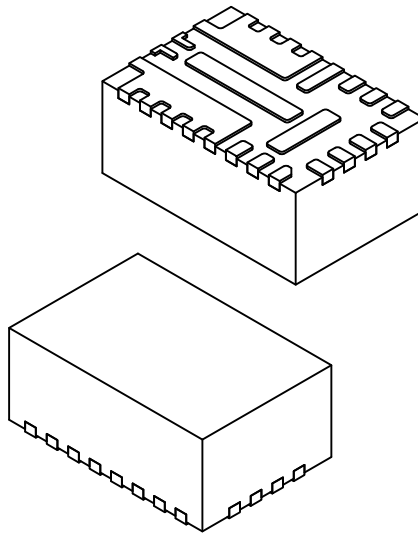
24-Lead Plastic Quad Flat, No Lead Package (QRA) - 3x4.5 mm Body [QFN]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



24-Lead Plastic Quad Flat, No Lead Package (QRA) - 3x4.5 mm Body [QFN]

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Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	24		
Pitch	e	0.50 BSC		
Overall Height	A	1.80	1.85	1.90
Standoff	A1	0.00	0.02	0.05
Terminal Thickness	A3	0.203 REF		
Overall Length	D	3.00 BSC		
Exposed Pad Length	D2	0.338	0.388	0.438
Exposed Pad Length	D3	1.344	1.394	1.444
Exposed Pad Length	D4	xx	1.30 ??	xx
Overall Width	E	4.50 BSC		
Exposed Pad Width	E2	2.35	2.40	2.45
Exposed Pad Width	E3	0.326	0.376	0.426
Terminal Width	b	0.20	0.25	0.30
Terminal Width	b2	0.08	0.13	0.18
Terminal Length	L	0.35	0.40	0.45
Terminal Length Setback	(L2)	0.20	0.25	0.30
Terminal-to-Exposed-Pad	K1	0.20	—	—
Terminal-to-Exposed-Pad	K2	0.20	—	—

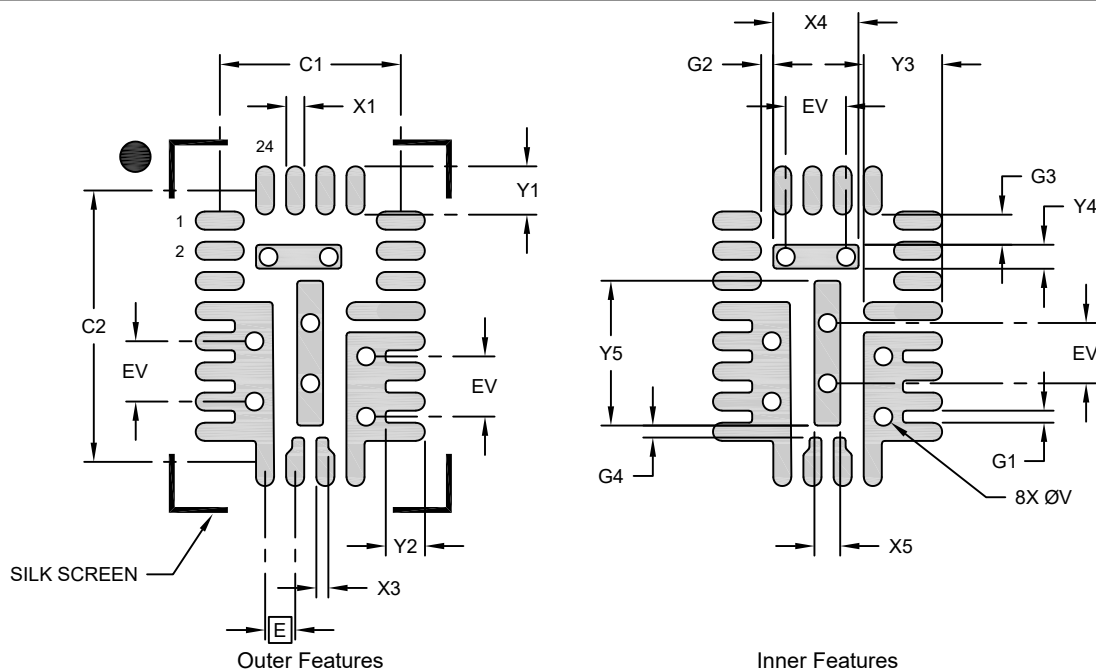
Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

24-Lead Plastic Quad Flat, No Lead Package (QRA) - 3x4.5 mm Body [QFN]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Contact Pad Spacing	C1			3.00
Contact Pad Spacing	C2			4.50
Contact Pad Width (X24)	X1		0.30	
Contact Pad Length (X23)	Y1		0.80	
Contact Pad Length Setback (X7)	Y2		0.65	
Contact Pad Width (X2)	X3		0.20	
Contact Pad Length (X1)	Y3		1.30	
Exposed Pad Length	X4			1.41
Exposed Pad Width	Y4			0.40
Exposed Pad Width	X5			0.43
Exposed Pad Length	Y5			2.40
Terminal to Terminal	G1	0.20		
Terminal to Exposed Pad	G2	0.20		
Terminal to Exposed Pad	G3	0.50		
Terminal to Exposed Pad	G4	0.20		
Thermal Via Diameter	V		0.30	
Thermal Via Pitch	EV		1.00	

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-3295 Rev A